

uSense tactile data description

Overview

This document summarizes key information about the data used in uSense.

Sensor Setup

- A 3×3 array of piezoresistive taxels (9 channels total) mounted on a 3D-printed finger.
- The finger swipes across physical textures at varying speeds and forces.
- Each taxel records voltage data corresponding to pressure changes during contact.
- A load cell is used underneath the texture to measure applied force.

Trial information

- Signals are sampled at approximately 338 Hz.
- Trial timeline:
 - At a certain time (the same for all speeds) the sensor touches the textures.
 - At a certain time (the same for all speed) the sensor swipes across the texture.
 - Next, initial contact begins (detected via load cell signal).
 - texture interaction duration varies depending on swipe speed.
- There are 21 unique textures.
- Each texture is recorded under different combinations of:
 - 5 swiping speeds: 2–10 cm/s
 - 4 pseudo forces: 0.0–1.5 mm displacement (i.e., controlled indentation depth)
 - 100 trials per condition $\Rightarrow$ 2,000 trials per texture
- A Total of 42,000 trials collected.

Texture encoding

Each trial file follows a structured naming format that specifies the texture index, swipe speed, applied force, and trial index:

[Data/Texture001/Data\[Texture\(001\)\]\[Speed\(1200\)\]\[Force\(0.0\)\]\[Trial\(1\)\].csv](#)

Swipe speeds are specified in mm/min and take values

$\{1200, 2400, 3600, 4800, 6000\}$,

which correspond to

$$\{2, 4, 6, 8, 10\} \text{ cm/s.}$$

Applied force (controlled indentation depth) takes values

$$\{0.0, 0.5, 1.0, 1.5\}.$$

The **Trial(k)** field denotes the repetition index under a fixed texture, speed, and force condition. **Texture0XY**, where:

- X indicates the texture geometric pattern:
 - 0: Smooth flat surface (control)
 - 0: Half-cylinder bumps
 - 1: Half-rectangular ridges
 - 2: Hemispherical (3D round) bumps
 - 3: Cone-shaped (3D pointed) bumps
- Y indicates bump density:
 - 1: 4 rows
 - 2: 5 rows
 - 3: 6 rows
 - 4: 3 rows
 - 5: 2 rows (least dense)

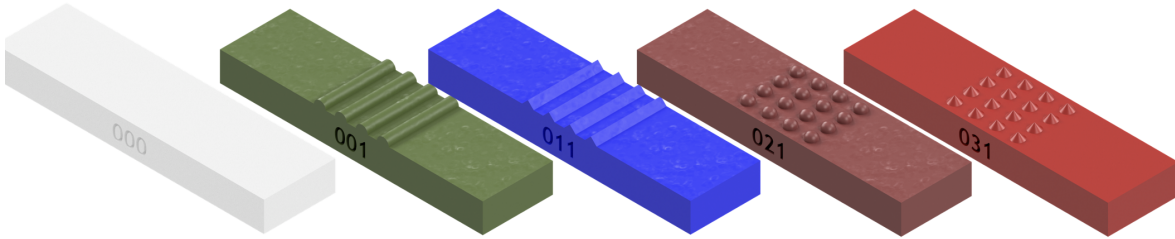


Figure 1: Representative physical textures used for tactile data collection. Labels follow the **Texture0XY** format, where X specifies the geometric pattern and Y specifies the bump density.

Data structure

- Each trial's data consists of an array of shape **(samples, 11)**:
 - Columns 1–9: raw voltage from each taxel
 - Column 10: load cell signal
 - Column 11: timestamp